GNN-Based BERT for Understanding Context from Music

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***Abstract*—Understanding musical context requires combining information from audio structure and natural-language descriptions. This project develops and evaluates a staged GNN–BERT framework for music understanding using four tasks. Task 1 establishes a BERT-based multi-label text baseline on MagnaTagATune using filename-derived text as a weak textual signal. Task 2 compares two audio representations on the same tracks: a two-layer GraphSAGE model operating on five-segment chroma graphs. Task 3 then combines MusicCaps audio graphs and captions using a GNN–BERT cross-attention model for 30-tag multi-label prediction. An ablation study shows that BERT-only performs slightly better than the proposed fusion. Lastly, Task 4 further learns a shared 128-dimensional audio–text embedding space using bidirectional contrastive learning.Overall, the experiments demonstrate that combining graph and language representations is feasible, but the current coarse graph representation and limited cross-modal alignment prevent the fusion and contrastive models from consistently outperforming simpler text-based alternatives.**

***Index Terms*—GNN, BERT, GraphSAGE, cross-attention, contrastive learning, music information retrieval.**

I. Introduction

Music contains several forms of information at the same time. A recording has acoustic properties such as timbre, rhythm, harmony, and instrumentation, while descriptions written by listeners can express concepts such as mood, genre, recording quality, or the presence of particular instruments. A useful music-understanding system therefore needs to capture both the structure of the audio and the meaning expressed in language.

A Graph Neural Network (GNN) provides one way to represent the structural side of music. In this project, each track is divided into a small number of segments, represented as graph nodes, while connections describe relationships between those segments. GraphSAGE then propagates information between connected nodes to produce a graph-level representation. This representation is useful for testing whether a structured view of an audio signal can support music classification. However, the choice of graph features is important because a very small graph can lose acoustic information that is present in a full spectrogram.

BERT provides the language side of the system. It converts a sequence of words into contextual representations in which each word is interpreted using its surrounding text. This makes it suitable for captions and tag-related descriptions. The project first evaluates BERT independently and then combines it with the graph representation so that information from the audio and text modalities can interact.

The project follows a four-task progression. Task 1 establishes a text-only BERT baseline for multi-label music-tag prediction. Task 2 establishes two audio-only baselines, comparing GraphSAGE on five-node chroma graphs with a CNN on mel-spectrograms using the same FMA-small tracks. Task 3 introduces GNN–BERT cross-attention on MusicCaps, where the graph representation is used to query BERT token representations before classification. The task also includes BERT-only, GNN-only, and early-concatenation ablations to determine whether cross-attention provides an advantage. Task 4 moves from supervised classification to cross-modal contrastive learning, training separate graph and text encoders to place matching audio and captions close together in a shared embedding space and evaluating the result through retrieval and zero-shot tagging.

The main purpose of this staged design is not only to build a multimodal model, but also to identify which parts of the representation contribute useful information. The results therefore provide a direct comparison between text-only, graph-only, spectrogram-based, supervised fusion, and contrastive approaches.

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II. Related Work

Understanding music computationally has been approached from several different modeling angles, each suited to a different aspect of what ”context” means for a piece of music.

Early work on sequential audio and music understanding relied on recurrent architectures or RNNs and LSTMs which process a track frame by frame and carry a hidden state forward through time. These models are a natural fit for signals like spectrograms, where the order of events matters and later frames depend on earlier ones. More recently, convolutional networks operating directly on spectrogram-like time-frequency representations have become a standard, strong baseline for tasks such as genre and tag classification: a spectrogram is treated much like an image, and the convolutional filters learn to detect local timbral and rhythmic patterns regardless of exactly where in the clip they occur. Both families share a common limitation, however: they reason over the signal as a flat sequence or grid, without an explicit notion of which parts of a track relate to which other parts beyond simple temporal proximity.

Graph-based approaches address this limitation directly. Representing a track as a graph with nodes standing in for segments, chords, or other musical events, and edges standing in for temporal adjacency, harmonic transition, or acoustic similarity allows a Graph Neural Network to propagate information between parts of a track that are musically related but not necessarily adjacent in time, such as a repeated chorus or a recurring motif. Message-passing architectures of this kind have been applied broadly to structured, non-grid data, and music structure graphs are a natural instance of this setting: the graph encodes a hypothesis about which parts of a track are related, and the network's job is to aggregate information along exactly those connections. This flexibility comes at a cost, however: a graph representation is only as informative as the features attached to its nodes and the edges chosen to connect them, and a coarse graph can discard information that a denser, grid-based representation would retain 2̌014 a trade-off this project's own Task 2 comparison speaks to directly.

On the textual side, transformer-based language models most notably BERT and its derivatives have become the standard tool for extracting contextual, semantically rich representations from short text, including the tags, captions, and lyrics associated with a piece of music. Unlike earlier bag-of-words or embedding-averaging approaches, a transformer encoder builds a representation of each token that already accounts for the surrounding words, which matters for music tags and captions where meaning is often compositional (e.g., ”slow acoustic ballad” versus ”fast electronic dance track” sharing individual words but describing very different music).

A smaller but growing body of work has begun to combine these two families structural graph encoders and semantic text encoders under the observation that neither modality alone fully captures what a listener means by musical context. Fusing a graph-level audio representation with a text-level semantic representation, whether through simple concatenation or a more expressive mechanism such as cross-attention, is a natural way to let the two signals inform each other rather than being modeled in isolation. This project follows that general direction: rather than committing immediately to a fused architecture, it first establishes how much each modality can explain on its own a text-only BERT baseline (Task 1) and two audio-only baselines contrasting graph-structured and grid-structured representations of the same tracks (Task 2) before supervised fusion and contrastive cross-modal learning are introduced.

TABLE I
Dataset Summary

Dataset	Raw Size	Used / Processed	Role
FMA-small	8,000 clips, 8 genres	7,994 segment graphs and 7,994 matched mel-spectrograms	Task 2: genre classification
MagnaTagATune	25,863 clips, 188 tags	Top-50 tags across all 25,863 clips	Task 1: BERT tag classification
MusicCaps	5,521 captioned clips	1,159 clips matched with locally available graph audio	Tasks 3–4: fusion, retrieval, and zero-shot tagging

III. Dataset and Preprocessing

Three datasets are used across the project, with each dataset serving a different role. FMA-small provides the audio data used to construct segment-level graphs and mel-spectrograms for Task 2. MagnaTagATune provides multi-label music tag data for the text-based BERT baseline in Task 1. MusicCaps provides paired audio and natural-language captions for the multimodal fusion and contrastive learning experiments in Tasks 3 and 4. A summary of the datasets is presented in Table I.

A. FMA-Small: Audio-to-Chroma-to-Graph Representation

FMA-small consists of 30-second audio clips sampled at 22,050 Hz. Each audio clip is divided into five fixed 5-second segments. For every segment, a 12-dimensional chroma feature vector is extracted using the short-time Fourier transform based chroma representation provided by `librosa.feature.chroma_stft`. The feature values are mean-pooled over time, producing a node feature matrix of size $(5, 12)$ for each track.

A graph is then constructed using the five segment nodes. Two types of connections are used. First, temporal adjacency edges connect consecutive segments $(i, i+1)$, preserving the original temporal order of the audio. Second, similarity edges connect each segment to its most cosine-similar non-adjacent segment. Both edge types are represented as bidirectional edges.

This process produces a compact five-node graph for every track. The resulting graph is stored using the PyTorch Geometric `Data(x, edge_index)` representation. A resumable preprocessing pipeline was used to avoid repeating successfully processed tracks after an interruption.

Of the 8,000 FMA-small clips attempted, 7,994 were successfully converted into graphs. The remaining 18 clips were excluded because they were too short to produce the required

TABLE II
MAGNATAGATUNE SPLIT SIZES FOR TASK 1

Split	Clips	Group Overlap
Train	20,558	0
Validation	2,726	0
Test	2,579	0

segments. The resulting dataset is approximately balanced across the eight genres: Pop, Folk, Instrumental, and International contain 1,000 clips each; Rock, Experimental, and Electronic contain 999 clips each; and Hip-Hop contains 997 clips.

B. MagnaTagATune: Tag Data and Pseudo-Text Construction

MagnaTagATune contains 25,863 labeled clips across 188 tags. For Task 1, the dataset is restricted to the 50 most frequent tags. Since MagnaTagATune does not provide natural-language captions, a pseudo-text representation is constructed from each audio filename.

The frame-range suffix in each filename, such as `-30-59`, is removed, and remaining hyphens and underscores are replaced with spaces. For example, a filename such as `artist-album-track-30-59.mp3` is converted into a text sequence similar to "artist album track". This provides a weak textual signal that can be processed by BERT.

The preprocessing also accounts for redundancy in MagnaTagATune. Multiple approximately 29-second clips may originate from the same underlying track and therefore share the same cleaned text string. The 25,863 clips correspond to only 5,405 unique cleaned text strings, with the most repeated string occurring 75 times.

To prevent the same underlying track from appearing in multiple dataset partitions, `GroupShuffleSplit` is applied using the cleaned text string as the grouping key. An 80/10/10 train-validation-test split is used, and the resulting groups have zero overlap between the three partitions. The resulting split sizes are shown in Table II.

The cleaned text is tokenized using the `bert-base-uncased` tokenizer. Padding and truncation are applied with a maximum sequence length of 128 tokens. The 50 tag columns are converted into a multi-hot floating-point label vector because a single clip may contain multiple tags.

C. MusicCaps: Caption and Graph Preparation

MusicCaps is used for the multimodal experiments in Tasks 3 and 4. The official caption release contains 5,521 captioned clips. From these, 1,159 clips could be matched with locally available audio and graph representations used by the project.

Because MusicCaps clips are approximately 10 seconds long, the audio is divided into five 2-second segments. Each segment is represented using the same 12-dimensional chroma feature representation used for the FMA-small graph pipeline. Consequently, each MusicCaps example is represented as a five-node graph, maintaining the same graph structure across the experiments.

The corresponding natural-language captions are tokenized using the `bert-base-uncased` tokenizer with a maximum sequence length of 128 tokens. The matched graph-caption pairs are divided into 927 training examples, 116 validation examples, and 116 test examples.

D. Mel-Spectrogram Extraction

A second audio representation is prepared for the Task 2 CNN baseline. For every FMA-small track, a log-mel spectrogram is extracted using `librosa.feature.melspectrogram` with 128 mel-frequency bins and the native 22,050 Hz sampling rate. The resulting power spectrogram is converted to decibel scale using `librosa.power_to_db`.

Since the exact number of time frames can vary slightly between tracks, each spectrogram is padded or truncated to 1,300 time frames. Therefore, every track is represented by a fixed-size $(128, 1300)$ spectrogram.

Of the 8,000 clips attempted for spectrogram extraction, 7,997 were successfully converted. After matching these results with the genre-labeled graph dataset, exactly 7,994 tracks contain both a valid graph and a valid mel-spectrogram. The identical set of 7,994 tracks is therefore used for both the GraphSAGE and CNN models, enabling a controlled comparison between the two audio representations.

IV. METHODOLOGY

The project follows a four-task progression, beginning with independent text and audio baselines and then moving toward multimodal fusion and cross-modal representation learning. The four tasks are designed to examine the contribution of text representations, graph-based audio representations, spectrogram-based representations, and multimodal learning.

A. Task 1: BERT Baseline for Music Tag Understanding

Model. The first task establishes a text-only baseline using the `bert-base-uncased` model. BERT contains 12 transformer layers with a hidden representation size of 768. The pooled `[CLS]` representation is passed through a linear classification head to produce predictions for the 50 music tags.

For an input text sequence X_{text}, the BERT representation and tag predictions are defined as

$$t = \mathrm{BERT}_{\mathrm{CLS}}(X_{\text{text}}), \qquad \hat{y}_k = \sigma(w_k^T t + b_k), \quad k = 1, \ldots, 50. \tag{1}$$

Since a clip may contain multiple tags simultaneously, the task is treated as a multi-label classification problem. Binary cross-entropy is therefore used for each tag:

$$L_{\mathrm{BERT}} = -\frac{1}{K}\sum_{k=1}^{K}\left[y_k \log(\hat{y}_k) + (1-y_k)\log(1-\hat{y}_k)\right], \quad K = 50. \tag{2}$$

The implementation uses `BCEWithLogitsLoss`, which combines the sigmoid operation and binary cross-entropy in a numerically stable manner.

The complete BERT encoder is fine-tuned end-to-end. AdamW is used as the optimizer with a learning rate of 2×10^{-5}. The batch size is 16 and training is performed for five epochs. During evaluation, sigmoid outputs are thresholded at 0.5 to obtain the predicted tag set. Macro-F1 and micro-F1 are calculated on the validation and test sets.

B. Task 2: GraphSAGE and CNN Audio Baselines

Task 2 compares two different representations of the same FMA-small audio tracks. The first model uses a graph representation and GraphSAGE, while the second uses the complete mel-spectrogram representation and a convolutional neural network. Both models use the same 7,994 tracks and the same train-validation-test partition.

1) GraphSAGE Model: The GraphSAGE model operates on the five-node chroma graph described in Section III. Each node initially contains a 12-dimensional chroma feature vector. Two GraphSAGE layers are used, increasing the feature dimension from 12 to 64 and maintaining 64 dimensions in the second layer.

For a node i at layer l, neighborhood information is aggregated using mean pooling:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \mathrm{CONCAT}\left(h_i^{(l)}, \underset{j\in\mathcal{N}(i)}{\mathrm{MEAN}} h_j^{(l)}\right)\right). \quad (3)$$

After the final GraphSAGE layer, a graph-level representation is obtained by mean-pooling all node embeddings:

$$g = \frac{1}{|V|}\sum_{i\in V} h_i^{(L)}, \qquad \hat{y} = \sigma(Wg + b). \quad (4)$$

The final linear layer maps the 64-dimensional graph representation to eight genre classes. Since each track has one genre label, categorical cross-entropy is used as the loss function. The model is trained using Adam with a learning rate of 0.001, a batch size of 64, and 15 training epochs.

2) CNN Baseline: The second audio model is a compact CNN that operates directly on the $(128, 1300)$ log-mel spectrograms. It contains three convolutional blocks with channel sizes $1 \to 16 \to 32 \to 64$. Each convolution uses a 3×3 kernel with padding of one, followed by a ReLU activation and 2×2 max-pooling.

After the convolutional blocks, adaptive average pooling reduces the feature map to a fixed 1×1 representation. A final linear layer maps the resulting 64-dimensional representation to the eight genre classes.

The CNN uses categorical cross-entropy and Adam with a learning rate of 0.001. A batch size of 32 is used because the spectrogram representation requires substantially more memory than the five-node graph. The model is trained for 15 epochs.

Table III summarizes the controlled comparison between GraphSAGE and the CNN.

The use of identical tracks and identical data splits allows the experiment to focus primarily on the difference between the graph-based chroma representation and the denser time-frequency mel-spectrogram representation.

TABLE III
TASK 2 BASELINE COMPARISON

Property	**GraphSAGE**	**CNN**
Input per track	5-node graph, 12-dim chroma vector per node	128×1300 log-mel spectrogram
Encoder	2-layer SAGEConv ($12 \to 64 \to 64$), mean readout	3-layer Conv2D ($1 \to 16 \to 32 \to 64$), global average pooling
Loss	Cross-entropy (8 classes)	Cross-entropy (8 classes)
Optimizer	Adam, $lr = 0.001$	Adam, $lr = 0.001$
Batch size	64	32
Epochs	15	15
Train / Val / Test	6,395 / 799 / 800	6,395 / 799 / 800

C. Task 3: GNN–BERT Cross-Attention Fusion

Task 3 combines the graph and language modalities for supervised multi-label music-tag prediction. Each MusicCaps example contains a five-node audio graph and its corresponding natural-language caption.

The graph branch uses two GraphSAGE layers followed by global mean pooling. The input node features have dimension 12 and are transformed into 64-dimensional node representations. Mean pooling produces a single 64-dimensional graph vector g.

The text branch uses `bert-base-uncased`. Unlike Task 1, where only the pooled `[CLS]` representation is used, Task 3 retains the complete sequence of contextual token representations.

Cross-attention is then used to allow the audio representation to select relevant information from the caption. The graph representation is projected to form the query Q, while the BERT token representations are projected to form the keys K and values V. The attention scores are computed using scaled dot-product attention:

$$A = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \quad (5)$$

where d is the dimensionality of the projected attention space.

The attended text representation is calculated as

$$z_{\text{text}} = AV. \quad (6)$$

The graph representation and attended text representation are then concatenated and passed to a linear classification layer:

$$z = \mathrm{CONCAT}(g, z_{\text{text}}), \qquad \hat{y} = Wz + b. \quad (7)$$

The classifier produces 30 tag logits. Since multiple tags may be correct for a single example, `BCEWithLogitsLoss` is used. The model is optimized using AdamW with a learning rate of 2×10^{-5} and a batch size of 8.

The complete implemented model contains 110,748,190 trainable parameters. The dataset consists of 927 training examples, 116 validation examples, and 116 test examples. Validation macro-F1 and micro-F1 are monitored during training, and the best validation model is retained.

To determine whether cross-attention provides an actual benefit, three ablation models are evaluated: BERT-only, GNN-only, and an early-concatenation model. The BERT-only model uses only the caption representation, the GNN-only model uses only the graph representation, and the early-concatenation model joins the two representations before classification.

D. Task 4: Contrastive Audio–Caption Learning

Task 4 changes the objective from supervised tag classification to cross-modal representation learning. A graph encoder and a text encoder are trained jointly so that matching audio–caption pairs obtain similar embeddings while mismatched pairs are pushed apart.

The graph encoder consists of two GraphSAGE layers followed by mean pooling and a projection layer that maps the resulting 64-dimensional graph representation into a 128-dimensional embedding space. The text encoder uses BERT and projects the 768-dimensional `[CLS]` representation into the same 128-dimensional space.

Both graph and text embeddings are L2-normalized before similarity calculation. For a batch containing N graph–caption pairs, a similarity matrix is constructed by computing the dot product between every graph embedding and every caption embedding.

A temperature value of $\tau = 0.07$ is used to scale the similarities. The graph-to-text contrastive loss is defined as

$$L_{g\to t} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N}\exp(s_{ij}/\tau)}, \tag{8}$$

where s_{ij} represents the similarity between graph i and caption j.

A corresponding caption-to-graph loss is calculated in the opposite direction:

$$L_{t\to g} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N}\exp(s_{ji}/\tau)}. \tag{9}$$

The final bidirectional InfoNCE objective is the average of the two losses:

$$L_{\text{InfoNCE}} = \frac{1}{2}\left(L_{g\to t} + L_{t\to g}\right). \tag{10}$$

The graph and text encoders are optimized jointly using AdamW with a learning rate of 2×10^{-5} and a batch size of 16. Validation loss is used for model selection and early stopping.

After training, the shared embedding space is evaluated in two retrieval directions. In caption-to-audio retrieval, a caption is used to retrieve the most similar audio graphs. In audio-to-caption retrieval, an audio graph is used to retrieve the most similar captions. Recall@1, Recall@5, and Recall@10 are used to measure whether the correct paired item appears among the top K retrieved candidates.

The learned representation is also evaluated for zero-shot music-tag prediction. Text prompts of the form "this song is [tag]" are embedded using the text encoder, and their similarity to test graph embeddings is used to determine the predicted tags.

TABLE IV
TASK 1 TEST-SET PERFORMANCE (MAGNATAGATUNE, 50 TAGS)

Metric	Loss	Macro-F1	Micro-F1
Test set	0.1420	0.2120	0.3411

TABLE V
TASK 1 QUALITATIVE PREDICTIONS ON TEST CLIPS

Pseudo-text input	True tags	Predicted tags
dac crowell the sea and the sky 01 tidal motion	slow, strings, electronic, ambient, synth	ambient
dac crowell the sea and the sky 02 umi no kami ni kansha	(none)	(none)
kitka nectar 11 miskolc felol hidegen fuj a szel hungary	drums, female, singing	(none)
curandero aras 03 segue	(none)	guitar
shane jackman sanctuary 05 simply put	guitar, male, male vocal, male voice	guitar

V. RESULTS AND ANALYSIS

A. Task 1: BERT Baseline for Music Tag Understanding

The fine-tuned `bert-base-uncased` tag classifier was evaluated on the held-out, track-grouped test split of 2,579 MagnaTagATune clips (Section **??**). Validation macro-F1 improved steadily across all 5 training epochs, from 0.1421 at epoch 1 to 0.2153 at epoch 5, while validation loss decreased from 0.1439 to 0.1345 over the same period. Table IV reports the final test-set performance.

The gap between macro-F1 (0.212) and micro-F1 (0.341) indicates the model performs considerably better on frequent tags than on rare ones, which is expected under a 50-way multi-label setup with a long-tailed tag distribution. Table V lists 5 randomly sampled test predictions, using the pseudo-text input described in Section **??** rather than genuine captions.

These examples illustrate both the model's partial competence and the ceiling imposed by the pseudo-text input: in the second row, an uninformative filename correctly yields no predicted tags; in the third row, a filename containing no lexical cues about instrumentation or vocals misses all three true tags; and in the fourth row, the model over-predicts `guitar` from a filename with no genuine tag signal at all. Overall, the model recovers tags that happen to be lexically present in the cleaned filename (e.g. inferring `guitar` when a track or album title plausibly references it) but cannot recover tags describing purely acoustic properties (e.g. `drums`, `female`) that the filename never mentions. This is consistent with the design limitation noted in Section **??**: the pseudo-text is a proxy, not a description of musical content, and macro-F1 of 0.212 should be read as the performance ceiling of that proxy rather than of BERT-based tagging in general.

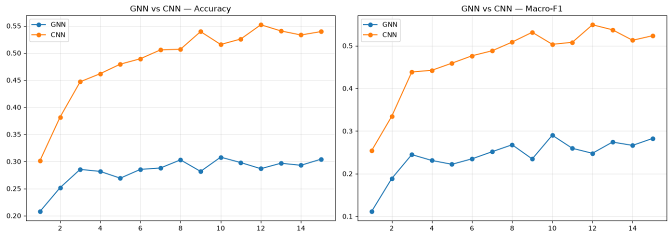

Fig. 1. Validation accuracy and macro-F1 for the GraphSAGE and CNN models across training epochs (Task 2).

TABLE VI
TASK 2 TEST-SET PERFORMANCE (FMA-SMALL, 8 GENRES)

Model	Loss	Accuracy	Macro-F1
GraphSAGE (graph)	1.8628	27.37%	0.2455
CNN (spectrogram)	1.3399	53.75%	0.5407

B. Task 2: GraphSAGE and CNN Baselines on FMA

Both Task 2 models were trained and evaluated on the identical 7,994-track FMA-small split (6,395 / 799 / 800). The GraphSAGE model's validation accuracy rose from 18.7% (epoch 1) to a peak of 30.2% (epoch 9), settling at 30.0% by epoch 15 with no clear further improvement in the final epochs. The CNN's validation accuracy rose more steeply and to a substantially higher ceiling, from 34.3% (epoch 1) to 56.3% (epoch 14), the best epoch observed. Fig. 1 plots both validation curves side by side.

Table VI reports final test-set performance for both models under identical evaluation conditions.

The CNN outperforms the GraphSAGE model by roughly a factor of two on both accuracy and macro-F1, despite both models sharing the same loss function, optimizer, learning rate, and - most importantly - the exact same 7,994 tracks and train/validation/test split (Table II). Since the comparison is controlled for everything except the input representation, the most defensible explanation is that reducing each 30-second track to five 12-dimensional chroma vectors discards genre-relevant information - likely timbral and textural detail that a full-resolution $128 \times 1{,}300$ mel-spectrogram retains and that a convolutional filter bank can exploit directly. Both models remain well above the 12.5% random-guess floor for an 8-class problem, so the graph-based model has learned a genuine, if weak, signal; it is simply working from a much coarser representation of the same audio.

C. Task 3: GNN–BERT Fusion and Ablations on MusicCaps

Task 3 uses the 1,159 MusicCaps clips with locally available audio, split 927 / 116 / 116, with the 30 most frequent free-text `aspect_list` phrases (e.g. *low quality*, *instrumental*, *emotional*, *passionate*) used as multi-label targets. The full cross-attention fusion model (110,748,190 parameters) was trained for 13 epochs; validation macro-F1 rose steadily from 0.0369 (epoch 1) to a peak of 0.6293 (epoch 12), with test-set evaluation performed on the final-epoch weights. Fig. 2 shows the full validation F1 trajectory.

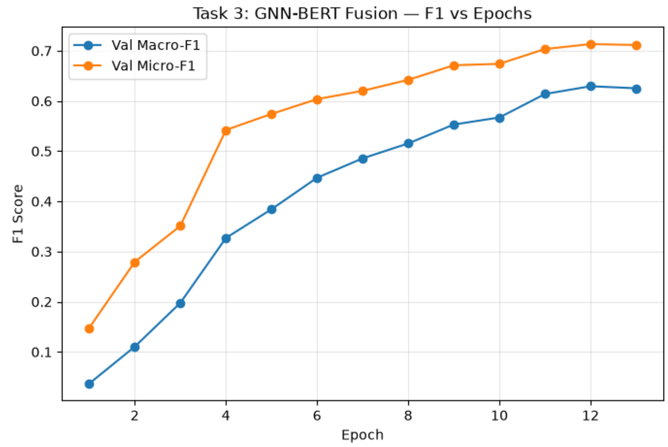

Fig. 2. Validation macro- and micro-F1 across training epochs for the GNN–BERT cross-attention fusion model (Task 3).

TABLE VII
TASK 3 ABLATION RESULTS (MUSICCAPS TEST SET, 30 TAGS)

Model	Test Macro-F1	Test Micro-F1
BERT-only	**0.5522**	**0.7159**
GNN-only	0.0000	0.0000
Early-Concat	0.5330	0.7123
Cross-Attention (Fusion)	0.5329	0.7007

1) Ablation Comparison: Three ablations - `BertOnlyAblation`, `GNNOnlyAblation`, and `EarlyConcatAblation` - were trained with the same shared training function, early stopping (patience 3 on validation loss), and best-checkpoint restoration, then compared against the full cross-attention fusion model on the same test split. Table VII reports all four.

Two results here run counter to what a fusion-motivated project would hope to find, and both are reported as observed rather than adjusted for. First, the `GNN-only` ablation collapses completely: it predicts zero positive tags for every single test clip (0 of 248 true positives recovered, with sigmoid outputs confined to a narrow 0.114–0.357 range, never crossing the 0.5 threshold), yielding a macro- and micro-F1 of exactly 0.0. Unlike Task 2, where the same segment-graph representation at least out-performed random guessing on genre, it carries no usable signal for MusicCaps' free-text aspect tags - most of which (e.g. *low quality*, *emotional*, *amateur recording*) describe production quality or subjective affect rather than the kind of chroma-based harmonic content a 5-node, 12-dimensional segment graph can represent. Second, neither fusion strategy improves on the `BERT-only` baseline: Early-Concat (0.5330) and Cross-Attention (0.5329) both score marginally *below* BERT-only (0.5522) on macro-F1, and similarly on micro-F1. This is consistent with the first finding - a branch that carries no standalone signal is unlikely to contribute positively once fused, and may introduce optimization noise instead. Taken together, these results suggest the bottleneck in Task 3 is not the fusion mechanism (cross-attention vs. concatenation makes almost no difference here)

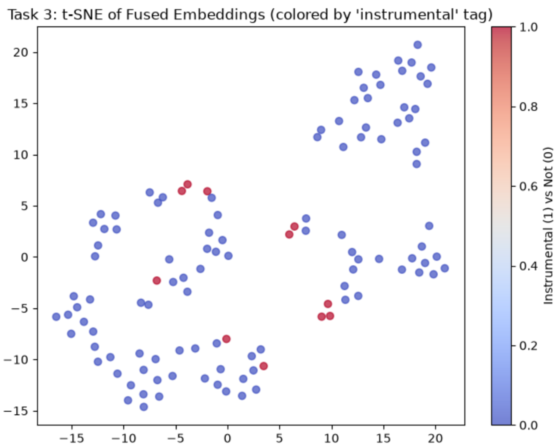

Fig. 3. t-SNE projection of fused GNN–BERT embeddings on the Task 3 test set, colored by the *instrumental* tag.

but the graph branch's input representation, echoing the Task 2 finding in Section V-B.

2) Embedding Visualization: Fig. 3 shows a t-SNE projection of the 832-dimensional fused embeddings ($z = \mathrm{CONCAT}(g, \text{attended\_text})$, 64 graph + 768 text dimensions) for all 116 test clips, colored by the binary *instrumental* tag.

3) Case Studies: Three test clips were examined in detail, inspecting the cross-attention weights over caption tokens alongside the true and predicted tag sets.

- **Clip 8UhdwnsckJ8.** Caption: *"The low quality recording features a tutorial that contains a flat male vocal talking over acoustic guitar strummed chords. The recording is noisy and in mono."* True tags: {low quality, noisy, mono}. Predicted tags: {low quality, noisy, mono} - an exact match. The top-5 attended tokens (`.`, `[CLS]`, *recording*, *mono*, *noisy*) do include two semantically relevant words (*mono*, *noisy*), though attention weights are all within 0.007, close to the $1/128 \approx 0.008$ uniform baseline for a 128-token sequence.
- **Clip 50fuQm8B2Yg.** Caption describes a jazz instrumental with saxophone, piano, and double bass. True tags: {instrumental, piano}. Predicted tags: {instrumental, slow tempo, piano} - both true tags recovered, plus one false positive (*slow tempo*). The top-5 attended tokens are all identical periods (`.`), each at weight 0.009, indicating the attention distribution here is essentially flat and not focused on any semantically informative word.
- **Clip B3lq6U4PDZo.** Caption describes an instrumental bell melody with a spooky theme. True tags: {instrumental}. Predicted tags: {instrumental} - an exact match. Attention weights are again nearly uniform (0.009–0.011 across the top-5 tokens, mostly periods and the word *this*).

Prediction accuracy on these three cases is encouraging, but the attention-weight pattern is a genuine limitation worth stating plainly: in all three cases, the cross-attention mechanism assigns near-uniform weight across caption tokens rather than sharply focusing on the words that determine the predicted tags. This was confirmed by re-computing attention with explicit padding-token masking (ruling out padding artifacts as the cause), which produced the same near-flat distribution. This suggests the fusion model's correct predictions in these cases may be driven more by the pooled graph embedding and the overall (unattended) text representation than by the cross-attention mechanism specifically learning *which* words to focus on - an important caveat given that interpretable graph-path/caption alignment was one of the motivations for using cross-attention over simple concatenation in the first place.

TABLE VIII
TASK 4 RETRIEVAL RESULTS (MUSICCAPS TEST SET, $N = 116$)

Direction	R1	R5	R10
Caption → Audio	1.72%	9.48%	20.69%
Audio → Caption	1.72%	8.62%	15.52%
Random chance	0.86%	4.31%	8.62%

D. Task 4: Contrastive GNN–BERT Alignment on MusicCaps

The dual-encoder contrastive model (a `GraphEncoder` and a `TextEncoder`, each projecting to a shared 128-dimensional, L2-normalized embedding space) was trained with the symmetric InfoNCE loss (temperature $\tau = 0.07$) on the same 1,159-clip MusicCaps split as Task 3. Training stopped early at epoch 7 (patience 3 on validation loss), with the best checkpoint from epoch 4 (validation loss 2.2889) restored for evaluation.

1) Retrieval Evaluation: Table VIII reports Recall@K in both retrieval directions on the 116-clip test set, alongside the random-chance baseline for a gallery of this size ($K/116$).

Both retrieval directions land at roughly 1.8–2.4$\times$ random chance across all three cutoffs, indicating the contrastive encoders have learned a real, if weak, alignment between graph and caption embeddings, rather than no alignment at all. In absolute terms, however, correctly retrieving the true match within the top 10 candidates out of 116 succeeds only 15–21% of the time, which is not yet a usable retrieval system.

2) Qualitative Retrieval Examples: Table IX lists all 10 required caption-to-audio retrieval examples on the test set.

Only 1 of the 10 examples (10%) retrieves the correct clip within the top 3, broadly consistent with the population-level R@5/R@10 figures in Table VIII. A more specific pattern is visible on inspection: the clips `9K8EePrEDdo`, `7S3fU4RHabw`, and `4ueN2gGsH5Y` appear as top-3 candidates for 7 of the 10 queries shown - including queries describing a saxophone jazz piece, a harmonica melody, a slide-guitar lesson, a synth drone, and a country vocal track, which have little in common. This is a hubness pattern common to under-trained contrastive embedding spaces, where a small number of embeddings sit close to the origin of the shared space and are returned as a nearest neighbor for many unrelated queries, rather than the model having learned genuinely query-specific alignment. This is consistent with the model being trained on

TABLE IX
TASK 4 QUALITATIVE RETRIEVAL EXAMPLES (CAPTION → AUDIO)

Query caption (truncated)	True ID	Top-3 retrieved	Hit
The low quality recording features a tutorial…	8UhdwnsckJ8	B3lq6U4PDZo, 4Mo7tdV2LZk, -Vo4CAMX26U	No
This jazz song features a saxophone…	50fuQm8B2Yg	9K8EePrEDdo, 7S3fU4RHabw, 7Vjp9y6wvkY	No
This instrumental song features a melody…	B3lq6U4PDZo	4ueN2gGsH5Y, 9K8EePrEDdo, 7S3fU4RHabw	No
This audio clip is a harmonica melody…	ApDJYsi9UGg	4ueN2gGsH5Y, 9K8EePrEDdo, 7S3fU4RHabw	No
The low quality recording features a harpsichord…	A_NkQM85g6Q	4ueN2gGsH5Y, 3XL825fK6UM, 9K8EePrEDdo	No
This is the recording of a slide guitar lesson…	19Pp9QEw17U	4ueN2gGsH5Y, 9K8EePrEDdo, 7S3fU4RHabw	No
A synth pad is playing a drone sound…	-Bu7YaslRW0	9K8EePrEDdo, 7S3fU4RHabw, 0lIETNZ_7sg	No
A male vocalist sings this mellow country song…	40xkia7fwEM	9K8EePrEDdo, 7S3fU4RHabw, 4ueN2gGsH5Y	No
The low quality recording features a live performance…	9K8EePrEDdo	4ueN2gGsH5Y, 3kSQNjfJnN8, 9K8EePrEDdo	**Yes**
This country song features a female voice…	75UH33tO0Bo	1dt9eL2rmSY, 8UhdwnsckJ8, 0KCVgexi4yU	No

TABLE X
ZERO-SHOT CONTRASTIVE TAGGING VS. TASK 3 SUPERVISED FUSION

Approach	Macro-F1	Micro-F1
Zero-shot (Task 4, contrastive)	0.0791	0.1215
Supervised (Task 3, fusion)	0.5329	0.7007

only 927 pairs for 7 effective epochs before early stopping, and suggests that more training data, longer training, or harder negative mining would likely be needed before retrieval quality is usable in practice.

3) Zero-Shot Tagging vs. Task 3 Supervised Fusion: Using the trained text encoder to embed 30 template phrases of the form *"this song is* $\{tag\}$*"*, cosine similarity between each test clip's graph embedding and each tag embedding was thresholded (mean + 1 standard deviation per clip) to produce zero-shot multi-label predictions. Table X compares this against the Task 3 supervised fusion model on the same test clips.

The supervised fusion model outperforms zero-shot tagging by roughly 6.7× on macro-F1 and 5.8× on micro-F1. This gap is expected in direction - a classifier trained directly on the tagging objective should outperform a similarity threshold applied to an embedding space optimized only for retrieval - but its size is also consistent with the weak retrieval alignment already observed in Table VIII: an embedding space that struggles to retrieve the single correct match for a caption cannot be expected to support fine-grained zero-shot tag thresholding. Zero-shot tagging from the contrastive model should accordingly be read as a proof of concept that the pipeline runs end-to-end, not yet as a competitive tagging method in its own right.

E. Synthesis Across Tasks

A consistent pattern emerges across all four tasks: every text-based or spectrogram-based component outperforms the segment-graph GNN branch built from 5-node, 12-dimensional chroma features, and every attempt to combine the graph branch with a stronger component (Task 3's fusion, Task 4's contrastive alignment) either fails to improve on the stronger component alone or produces only weak absolute performance. The GraphSAGE genre classifier in Task 2 (27.4% accuracy) is roughly half as accurate as the CNN baseline on identical data; the GNN-only ablation in Task 3 collapses entirely (0.0 macro-F1); and neither fusion strategy in Task 3 improves on BERT-only. This does not indicate that a graph-structured view of music is unhelpful in principle - Task 2's GraphSAGE model still clears the random-guess baseline by more than double, showing the graph does carry *some* genre-relevant signal - but it does indicate that, at the current 5-segment, 12-dimensional-chroma granularity, the graph branch is consistently the weakest link in this pipeline. The most promising next step, ahead of any further fusion-architecture tuning, is likely to enrich the segment graph itself: finer-grained segmentation, additional per-segment features beyond chroma (e.g. MFCCs, spectral contrast, or learned embeddings), or a larger receptive field via additional GraphSAGE layers.

TABLE XI
PERFORMANCE COMPARISON (EXPERIMENTAL RESULTS).

Model	Macro-F1	AUC-PR	R@5 (retrieval)
CNN mel-spec	0.5440	0.59863	–
Task 1: BERT-only	0.2057	0.32925	–
Task 2: GNN-only	0.2679	0.27262	–
Task 3: GNN-BERT	0.53287	0.6826	–
Task 4: Contrastive	0.079145	0.1203	0.0862

VI. CONCLUSION

This project investigated a staged GNN-BERT framework for understanding musical context by integrating structural audio graphs with semantic text representations. Through a four-task progression, individual baselines were established for text and audio modalities, followed by evaluations of multimodal fusion and cross-modal contrastive alignment.

The primary finding across all experiments is that while combining language and graph representations is conceptually sound, the model's overall effectiveness is heavily bottlenecked by the granularity of the input features. The results demonstrated that a coarse 5-node, 12-dimensional chroma graph discards crucial acoustic information, leading to significantly lower performance compared to dense, spectrogram-based CNNs and text-only BERT baselines. As a result, the supervised cross-attention fusion (Task 3) and contrastive representation learning (Task 4) struggled to outperform unimodal approaches, with the graph branch consistently providing an insufficient standalone signal.

Ultimately, for multimodal music models to effectively capture the complex, multi-layered nature of musical context, the structural audio representations must be substantially enriched. Future work should focus on improving the audio segment graph through finer-grained segmentation, the inclusion of denser acoustic features such as MFCCs, and expanded receptive fields within the Graph Neural Network architecture.

VII. References

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